

Earth Lighting Model

5.4

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Chapter 1

Module Index

1.1 Modules

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Chapter 2

Namespace Index

2.1 Namespace List

Here is a list of all namespaces with brief descriptions:

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Chapter 3

Data Structure Index

3.1 Data Structures

Here are the data structures with brief descriptions:

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Chapter 4

File Index

4.1 File List

Here is a list of all files with brief descriptions:

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Chapter 5

Module Documentation

5.1 Models

Modules

- [Environment](#)

5.1.1 Detailed Description

5.2 Environment

Modules

- [EarthLighting](#)

5.2.1 Detailed Description

5.3 EarthLighting

Files

- file [class_declarations.hh](#)
Forward declarations of classes defined for JEOD 2.0 Earth Lighting.
- file [earth_lighting.hh](#)
Calculates lighting information from the Earth, Sun and Moon for a vehicle in low Earth orbit.
- file [earth_lighting_messages.hh](#)
Implement earth_lighting_messages.
- file [earth_lighting.cc](#)
Implementation of the classes necessary for calculating Sun, Moon and Earth lighting effects on a vehicle orbiting the Earth.
- file [earth_lighting_messages.cc](#)
Implement earth_lighting_messages.

5.3.1 Detailed Description

Chapter 6

Namespace Documentation

6.1 jeod Namespace Reference

Namespace jeod.

Data Structures

- class [LightingBody](#)
Represents a major source of light in a space environment, such as the sun, the Earth, the moon, etc.
- class [LightingParams](#)
Contains important parameters for lighting information.
- class [EarthLighting](#)
A class for calculating lighting effects in low Earth orbit.
- class [EarthLightingMessages](#)
Describes messages used in the earth lighting model.

Variables

- static constexpr double [epsilon](#) = 1.0e-12

6.1.1 Detailed Description

Namespace jeod.

6.1.2 Variable Documentation

6.1.2.1 epsilon

```
constexpr double jeod::epsilon = 1.0e-12 [static], [constexpr]
```

Definition at line 56 of file earth_lighting.cc.

Referenced by jeod::EarthLighting::circle_intersect().

Chapter 7

Data Structure Documentation

7.1 jeod::EarthLighting Class Reference

A class for calculating lighting effects in low Earth orbit.

```
#include <earth_lighting.hh>
```

Public Member Functions

- [EarthLighting](#) ()=default
- [~EarthLighting](#) ()=default
- [EarthLighting](#) & [operator=](#) (const [EarthLighting](#) &)=delete
- [EarthLighting](#) (const [EarthLighting](#) &)=delete
- void [initialize](#) (DynManager &manager)
Initializes the [EarthLighting](#) object form the DynManager object.
- int [circle_intersect](#) (double r_bottom, double r_top, double d_centers, double *area)
- void [calc_lighting](#) (const double pos_veh[3])
Calculate earth lighting effects at the given position.

Data Fields

- bool [active](#) {true}
flag for if the model is active or not
- Planet * [earth](#) {}
Pointer to the Earth planet from the DynManager.
- Planet * [moon](#) {}
Pointer to the Moon planet from the DynManager.
- Planet * [sun](#) {}
Pointer to the Sun planet from the DynManager.
- const RefFrame * [earth_frame](#) {}
Pointer to the translation information for Earth inertial.
- const RefFrame * [moon_frame](#) {}
Pointer to the translation information for Moon inertial.
- const RefFrame * [sun_frame](#) {}
Pointer to the translation information for Sun inertial.

- [LightingBody sun_body](#)
Sun stellar parameters.
- [LightingBody earth_body](#)
Earth planetary parameters.
- [LightingBody moon_body](#)
Lunar planetary parameters.
- [LightingParams sun_earth](#)
Lighting of sun w.r.t.
- [LightingParams moon_earth](#)
Lighting of moon w.r.t.
- [LightingParams earth_albedo](#)
Earth albedo lighting.

Protected Attributes

- double [pos_moon](#) [3] {}
Moon position wrt Earth inertial.
- double [pos_sun](#) [3] {}
Sun position wrt Earth inertial.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) (jeod, [EarthLighting](#))

7.1.1 Detailed Description

A class for calculating lighting effects in low Earth orbit.

Definition at line 151 of file earth_lighting.hh.

7.1.2 Constructor & Destructor Documentation

7.1.2.1 [EarthLighting\(\)](#) [1/2]

```
jeod::EarthLighting::EarthLighting ( ) [default]
```

7.1.2.2 [~EarthLighting\(\)](#)

```
jeod::EarthLighting::~~EarthLighting ( ) [default]
```

7.1.2.3 EarthLighting() [2/2]

```
jeod::EarthLighting::EarthLighting (
    const EarthLighting & ) [delete]
```

7.1.3 Member Function Documentation

7.1.3.1 calc_lighting()

```
void jeod::EarthLighting::calc_lighting (
    const double pos_veh[3] )
```

Calculate earth lighting effects at the given position.

Parameters

in	<i>pos_veh</i>	The position of the point of interest in the earth inertial frame Units: M
----	----------------	---

Definition at line 233 of file `earth_lighting.cc`.

References `active`, `circle_intersect()`, `jeod::LightingBody::distance`, `earth_albedo`, `earth_body`, `earth_frame`, `jeod::LightingBody::half_angle`, `jeod::LightingParams::lighting`, `moon_body`, `moon_earth`, `moon_frame`, `jeod::LightingParams::obs_angle`, `jeod::LightingParams::occlusion`, `jeod::LightingParams::phase`, `pos_moon`, `pos_sun`, `jeod::LightingBody::position`, `jeod::LightingBody::radius`, `sun_body`, `sun_earth`, `sun_frame`, and `jeod::LightingParams::visible`.

7.1.3.2 circle_intersect()

```
int jeod::EarthLighting::circle_intersect (
    double r_bottom,
    double r_top,
    double d_centers,
    double * area )
```

Definition at line 148 of file `earth_lighting.cc`.

References `jeod::epsilon`.

Referenced by `calc_lighting()`.

7.1.3.3 initialize()

```
void jeod::EarthLighting::initialize (
    DynManager & manager )
```

Initializes the [EarthLighting](#) object form the DynManager object.

Will find the Earth, Sun and Moon objects and do the necessary setup to calculate earth lighting

Parameters

<code>in</code>	<code>manager</code>	The Dyn Manager object that includes the ephemeris for Sun, Earth and Moon
-----------------	----------------------	--

Definition at line 75 of file `earth_lighting.cc`.

References `earth`, `earth_body`, `earth_frame`, `jeod::EarthLightingMessages::initialization_error`, `moon`, `moon_body`, `moon_frame`, `jeod::LightingBody::radius`, `sun`, `sun_body`, and `sun_frame`.

7.1.3.4 operator=()

```
EarthLighting& jeod::EarthLighting::operator= (
    const EarthLighting & ) [delete]
```

7.1.3.5 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::EarthLighting::p (
    jeod ,
    EarthLighting ) [private]
```

7.1.4 Field Documentation**7.1.4.1 active**

```
bool jeod::EarthLighting::active {true}
```

flag for if the model is active or not

`trick_units(-)`

Definition at line 170 of file `earth_lighting.hh`.

Referenced by `calc_lighting()`.

7.1.4.2 earth

```
Planet* jeod::EarthLighting::earth {}
```

Pointer to the Earth planet from the `DynManager`.

`trick_units(-)`

Definition at line 175 of file `earth_lighting.hh`.

Referenced by `initialize()`.

7.1.4.3 earth_albedo

`LightingParams jeod::EarthLighting::earth_albedo`

Earth albedo lighting.

`trick_units(-)`

Definition at line 230 of file `earth_lighting.hh`.

Referenced by `calc_lighting()`.

7.1.4.4 earth_body

`LightingBody jeod::EarthLighting::earth_body`

Earth planetary parameters.

`trick_units(-)`

Definition at line 210 of file `earth_lighting.hh`.

Referenced by `calc_lighting()`, and `initialize()`.

7.1.4.5 earth_frame

```
const RefFrame* jeod::EarthLighting::earth_frame {}
```

Pointer to the translation information for Earth inertial.

`trick_units(-)`

Definition at line 190 of file `earth_lighting.hh`.

Referenced by `calc_lighting()`, and `initialize()`.

7.1.4.6 moon

```
Planet* jeod::EarthLighting::moon {}
```

Pointer to the Moon planet from the DynManager.

`trick_units(-)`

Definition at line 180 of file `earth_lighting.hh`.

Referenced by `initialize()`.

7.1.4.7 moon_body

`LightingBody jeod::EarthLighting::moon_body`

Lunar planetary parameters.

`trick_units(-)`

Definition at line 215 of file `earth_lighting.hh`.

Referenced by `calc_lighting()`, and `initialize()`.

7.1.4.8 moon_earth

`LightingParams jeod::EarthLighting::moon_earth`

Lighting of moon w.r.t.

`vehicle.trick_units(-)`

Definition at line 225 of file `earth_lighting.hh`.

Referenced by `calc_lighting()`.

7.1.4.9 moon_frame

`const RefFrame* jeod::EarthLighting::moon_frame {}`

Pointer to the translation information for Moon inertial.

`trick_units(-)`

Definition at line 195 of file `earth_lighting.hh`.

Referenced by `calc_lighting()`, and `initialize()`.

7.1.4.10 pos_moon

`double jeod::EarthLighting::pos_moon[3] {} [protected]`

Moon position wrt Earth inertial.

`trick_units(m)`

Definition at line 238 of file `earth_lighting.hh`.

Referenced by `calc_lighting()`.

7.1.4.11 pos_sun

```
double jeod::EarthLighting::pos_sun[3] {} [protected]
```

Sun position wrt Earth inertial.

trick_units(m)

Definition at line 243 of file earth_lighting.hh.

Referenced by calc_lighting().

7.1.4.12 sun

```
Planet* jeod::EarthLighting::sun {}
```

Pointer to the Sun planet from the DynManager.

trick_units(-)

Definition at line 185 of file earth_lighting.hh.

Referenced by initialize().

7.1.4.13 sun_body

```
LightingBody jeod::EarthLighting::sun_body
```

Sun stellar parameters.

trick_units(-)

Definition at line 205 of file earth_lighting.hh.

Referenced by calc_lighting(), and initialize().

7.1.4.14 sun_earth

```
LightingParams jeod::EarthLighting::sun_earth
```

Lighting of sun w.r.t.

vehicle. trick_units(-)

Definition at line 220 of file earth_lighting.hh.

Referenced by calc_lighting().

7.1.4.15 sun_frame

```
const RefFrame* jeod::EarthLighting::sun_frame {}
```

Pointer to the translation information for Sun inertial.

trick_units(-)

Definition at line 200 of file earth_lighting.hh.

Referenced by calc_lighting(), and initialize().

The documentation for this class was generated from the following files:

- [earth_lighting.hh](#)
- [earth_lighting.cc](#)

7.2 jeod::EarthLightingMessages Class Reference

Describes messages used in the earth lighting model.

```
#include <earth_lighting_messages.hh>
```

Public Member Functions

- [EarthLightingMessages](#) ()=delete
- [EarthLightingMessages](#) (const [EarthLightingMessages](#) &rhs)=delete
- [EarthLightingMessages](#) & operator= (const [EarthLightingMessages](#) &rhs)=delete

Static Public Attributes

- static const char * [initialization_error](#) = "environment/earth_lighting/" "initialization_error"
Indicates an error during initialization.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [EarthLightingMessages](#))

7.2.1 Detailed Description

Describes messages used in the earth lighting model.

Definition at line 83 of file earth_lighting_messages.hh.

7.2.2 Constructor & Destructor Documentation

7.2.2.1 EarthLightingMessages() [1/2]

```
jeod::EarthLightingMessages::EarthLightingMessages ( ) [delete]
```

7.2.2.2 EarthLightingMessages() [2/2]

```
jeod::EarthLightingMessages::EarthLightingMessages (
    const EarthLightingMessages & rhs ) [delete]
```

7.2.3 Member Function Documentation

7.2.3.1 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::EarthLightingMessages::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    EarthLightingMessages ) [private]
```

7.2.3.2 operator=()

```
EarthLightingMessages& jeod::EarthLightingMessages::operator= (
    const EarthLightingMessages & rhs ) [delete]
```

7.2.4 Field Documentation

7.2.4.1 initialization_error

```
char const * jeod::EarthLightingMessages::initialization_error = "environment/earth_lighting/"
"initialization_error" [static]
```

Indicates an error during initialization.

trick_units(-)

Definition at line 93 of file `earth_lighting_messages.hh`.

Referenced by `jeod::EarthLighting::initialize()`.

The documentation for this class was generated from the following files:

- [earth_lighting_messages.hh](#)
- [earth_lighting_messages.cc](#)

7.3 jeod::LightingBody Class Reference

Represents a major source of light in a space environment, such as the sun, the Earth, the moon, etc.

```
#include <earth_lighting.hh>
```

Public Member Functions

- [LightingBody](#) ()=default
- [~LightingBody](#) ()=default
- [LightingBody](#) & [operator=](#) (const [LightingBody](#) &)=delete
- [LightingBody](#) (const [LightingBody](#) &)=delete

Data Fields

- double [radius](#) {}
Celestial body mean equitorial radius.
- double [position](#) [3] {}
Inertial position relative to observer.
- double [distance](#) {}
Distance from observer to light body.
- double [half_angle](#) {}
Apparent half angle of body disk.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) (jeod, [LightingBody](#))

7.3.1 Detailed Description

Represents a major source of light in a space environment, such as the sun, the Earth, the moon, etc.

Definition at line 82 of file `earth_lighting.hh`.

7.3.2 Constructor & Destructor Documentation

7.3.2.1 LightingBody() [1/2]

```
jeod::LightingBody::LightingBody ( ) [default]
```

7.3.2.2 ~LightingBody()

```
jeod::LightingBody::~~LightingBody ( ) [default]
```

7.3.2.3 LightingBody() [2/2]

```
jeod::LightingBody::LightingBody (
    const LightingBody & ) [delete]
```

7.3.3 Member Function Documentation

7.3.3.1 operator=()

```
LightingBody& jeod::LightingBody::operator= (
    const LightingBody & ) [delete]
```

7.3.3.2 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::LightingBody::p (
    jeod ,
    LightingBody ) [private]
```

7.3.4 Field Documentation

7.3.4.1 distance

```
double jeod::LightingBody::distance {}
```

Distance from observer to light body.

trick_units(m)

Definition at line 103 of file earth_lighting.hh.

Referenced by jeod::EarthLighting::calc_lighting().

7.3.4.2 half_angle

```
double jeod::LightingBody::half_angle {}
```

Apparent half angle of body disk.

trick_units(rad)

Definition at line 108 of file earth_lighting.hh.

Referenced by jeod::EarthLighting::calc_lighting().

7.3.4.3 position

```
double jeod::LightingBody::position[3] {}
```

Inertial position relative to observer.

trick_units(m)

Definition at line 98 of file earth_lighting.hh.

Referenced by jeod::EarthLighting::calc_lighting().

7.3.4.4 radius

```
double jeod::LightingBody::radius {}
```

Celestial body mean equitorial radius.

trick_units(m)

Definition at line 93 of file earth_lighting.hh.

Referenced by jeod::EarthLighting::calc_lighting(), and jeod::EarthLighting::initialize().

The documentation for this class was generated from the following file:

- [earth_lighting.hh](#)

7.4 jeod::LightingParams Class Reference

Contains important parameters for lighting information.

```
#include <earth_lighting.hh>
```

Public Member Functions

- [LightingParams](#) ()=default
- [~LightingParams](#) ()=default
- [LightingParams](#) & [operator=](#) (const [LightingParams](#) &)=delete
- [LightingParams](#) (const [LightingParams](#) &)=delete

Data Fields

- double [obs_angle](#) {}
Apparent observation angle from light source.
- double [phase](#) {}
Apparent lighting phase of planet.
- double [occlusion](#) {}
Fraction of planetary surface occlusion.
- double [visible](#) {}
Fraction of planetary surface visible.
- double [lighting](#) {}
*Fraction of lighting (phase * visible).*

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) (jeod, [LightingParams](#))

7.4.1 Detailed Description

Contains important parameters for lighting information.

Definition at line 114 of file earth_lighting.hh.

7.4.2 Constructor & Destructor Documentation

7.4.2.1 LightingParams() [1/2]

```
jeod::LightingParams::LightingParams ( ) [default]
```

7.4.2.2 ~LightingParams()

```
jeod::LightingParams::~~LightingParams ( ) [default]
```

7.4.2.3 LightingParams() [2/2]

```
jeod::LightingParams::LightingParams (
    const LightingParams & ) [delete]
```

7.4.3 Member Function Documentation

7.4.3.1 operator=()

```
LightingParams& jeod::LightingParams::operator= (
    const LightingParams & ) [delete]
```

7.4.3.2 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::LightingParams::p (
    jeod ,
    LightingParams ) [private]
```

7.4.4 Field Documentation

7.4.4.1 lighting

```
double jeod::LightingParams::lighting {}
```

Fraction of lighting (phase * visible).

trick_units(—)

Definition at line 145 of file earth_lighting.hh.

Referenced by jeod::EarthLighting::calc_lighting().

7.4.4.2 obs_angle

```
double jeod::LightingParams::obs_angle {}
```

Apparent observation angle from light source.

trick_units(rad)

Definition at line 125 of file earth_lighting.hh.

Referenced by jeod::EarthLighting::calc_lighting().

7.4.4.3 occlusion

```
double jeod::LightingParams::occlusion {}
```

Fraction of planetary surface occlusion.

trick_units(—)

Definition at line 135 of file earth_lighting.hh.

Referenced by jeod::EarthLighting::calc_lighting().

7.4.4.4 phase

```
double jeod::LightingParams::phase {}
```

Apparent lighting phase of planet.

trick_units(—)

Definition at line 130 of file earth_lighting.hh.

Referenced by jeod::EarthLighting::calc_lighting().

7.4.4.5 visible

```
double jeod::LightingParams::visible {}
```

Fraction of planetary surface visible.

trick_units(—)

Definition at line 140 of file earth_lighting.hh.

Referenced by jeod::EarthLighting::calc_lighting().

The documentation for this class was generated from the following file:

- [earth_lighting.hh](#)

Chapter 8

File Documentation

8.1 `class_declarations.hh` File Reference

Forward declarations of classes defined for JEOD 2.0 Earth Lighting.

Namespaces

- [jeod](#)

Namespace jeod.

8.1.1 Detailed Description

Forward declarations of classes defined for JEOD 2.0 Earth Lighting.

8.2 `earth_lighting.cc` File Reference

Implementation of the classes necessary for calculating Sun, Moon and Earth lighting effects on a vehicle orbiting the Earth.

```
#include <cmath>
#include <cstdlib>
#include "dynamics/dyn_manager/include/dyn_manager.hh"
#include "environment/planet/include/planet.hh"
#include "utils/math/include/vector3.hh"
#include "utils/message/include/message_handler.hh"
#include "../include/earth_lighting.hh"
#include "../include/earth_lighting_messages.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

Variables

- static constexpr double `jeod::epsilon` = 1.0e-12

8.2.1 Detailed Description

Implementation of the classes necessary for calculating Sun, Moon and Earth lighting effects on a vehicle orbiting the Earth.

8.3 earth_lighting.hh File Reference

Calculates lighting information from the Earth, Sun and Moon for a vehicle in low Earth orbit.

```
#include "dynamics/dyn_manager/include/class_declarations.hh"
#include "environment/planet/include/class_declarations.hh"
#include "utils/ref_frames/include/class_declarations.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "environment/planet/include/planet.hh"
#include "utils/ref_frames/include/ref_frame.hh"
```

Data Structures

- class `jeod::LightingBody`
Represents a major source of light in a space environment, such as the sun, the Earth, the moon, etc.
- class `jeod::LightingParams`
Contains important parameters for lighting information.
- class `jeod::EarthLighting`
A class for calculating lighting effects in low Earth orbit.

Namespaces

- `jeod`
Namespace jeod.

8.3.1 Detailed Description

Calculates lighting information from the Earth, Sun and Moon for a vehicle in low Earth orbit.

8.4 earth_lighting_messages.cc File Reference

Implement earth_lighting_messages.

```
#include "utils/message/include/make_message_code.hh"
#include "../include/earth_lighting_messages.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

Macros

- `#define` [MAKE_EARTHLIGHTING_MESSAGE_CODE](#)(id) JEOD_MAKE_MESSAGE_CODE(EarthLightingMessages, "environment/earth_lighting/", id)

8.4.1 Detailed Description

Implement earth_lighting_messages.

8.4.2 Macro Definition Documentation

8.4.2.1 MAKE_EARTHLIGHTING_MESSAGE_CODE

```
#define MAKE_EARTHLIGHTING_MESSAGE_CODE(  
    id )  JEOD_MAKE_MESSAGE_CODE(EarthLightingMessages, "environment/earth_lighting/",  
    id)
```

Definition at line 43 of file earth_lighting_messages.cc.

8.5 earth_lighting_messages.hh File Reference

Implement earth_lighting_messages.

```
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::EarthLightingMessages](#)
Describes messages used in the earth lighting model.

Namespaces

- [jeod](#)
Namespace jeod.

8.5.1 Detailed Description

Implement earth_lighting_messages.

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